Synthetic cold-start splits are not enough: benchmarking temporal and provenance shifts in drug–target interaction prediction

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Abstract

Drug–target interaction (DTI) models are routinely validated on synthetic cold-start splits, yet practitioners deploy them under temporal and provenance shifts that these splits do not capture. Are these two evaluation settings interchangeable? We answer this question with a seven-model panel spanning descriptor-based (MLP, Random Forest), language-model (DTI-LM, HyperPCM), graph-based (GraphDTA-style), cross-attention (DrugBAN), and Transformer-fusion (ColdstartCPI-style) architectures, evaluated across three synthetic `BindingDB_Kd` cold-start splits, two temporal/provenance benchmarks (patent and non-patent BindingDB), and two supporting target-cold benchmarks (`BindingDB_Ki`, `DAVIS`). Three findings emerge. First, even within the synthetic panel the winner changes with the split definition. Second, temporal/provenance benchmarks produce a different ranking from the synthetic panel—and the non-patent temporal benchmark confirms that this is not a patent-provenance artifact. Third, overlap-restricted variants within the temporal family change the winner yet again, and an overlap-bucket diagnosis reveals that the temporal shift is structured rather than monolithic: different overlap strata favor different model families. These results demonstrate that synthetic cold-start and temporal/provenance benchmarks are non-interchangeable validation axes, and that benchmark-qualified reporting is necessary when making DTI generalization claims. We release reusable split manifests and a standalone evaluator to support reproducible comparison.

Keywords: drug–target interaction, benchmark design, temporal validation, distribution shift, BindingDB

1 Introduction

Predicting drug–target interactions (DTIs) for novel compounds or proteins is a central task in computational drug discovery. Because prospective wet-lab validation is expensive and slow, the field relies heavily on retrospective held-out evaluation. The most common protocol is *synthetic cold-start splitting*: compounds, targets, or both are held out at random so that the test set contains entities absent from training [3, 10, 14]. Under this protocol, dozens of recent architectures—from language-model pooling [1] to graph neural networks [13] to contrastive pre-training [18] and bilinear cross-attention [2]—have reported competitive or state-of-the-art results, often on a single split of a single dataset. The implicit assumption is that strong performance on one cold-start split implies robust generalization to deployment conditions.

This assumption, however, has not been directly tested against temporal and provenance shifts in DTI. In adjacent domains, mounting evidence shows that evaluation design can change model rankings: time-split cross-validation better predicts prospective performance than random splits in QSAR [17]; scaffold splits and random splits can disagree on the best molecular property predictor [22]; and domain-shifted evaluation alters conclusions in drug–drug interaction and molecular generalizability studies [4, 16]. Large-scale benchmarking efforts such as MoleculeNet [21] and the Therapeutics Data Commons [6] have further highlighted that no single model dominates across all tasks and splits. Yet in DTI specifically, the question of whether *synthetic cold-start rankings transfer to temporal/provenance-shifted settings* has not been posed as a controlled experiment.

We address this gap. Rather than proposing a new architecture, we construct a systematic benchmark study that directly compares model rankings across synthetic cold-start and temporal/provenance evaluation axes. We use a seven-model panel deliberately spanning the major architectural families in current DTI research: a descriptor-based MLP (`base`), a Random Forest on the same descriptors (`RF`), two language-model baselines (`DTI-LM` and `HyperPCM`), a graph-based baseline (`GraphDTA-style`), and a cross-attention baseline (`DrugBAN`). This panel is evaluated on seven benchmarks organized into three families: synthetic cold-start (`BindingDB_Kd`), temporal/provenance (patent and non-patent BindingDB time splits), and supporting target-cold OOD (`BindingDB_Ki`, `DAVIS`).

Our main contributions are:

1. **Ranking non-transferability.** We show that model rankings on synthetic cold-start splits do not transfer to temporal/provenance benchmarks: the identity of the best model changes across evaluation axes, and this is confirmed with paired bootstrap confidence intervals.
2. **Provenance separation.** By constructing both patent-derived and non-patent temporal benchmarks, we demonstrate that the ranking change is not merely a patent-provenance artifact—it persists on non-patent time-split data.
3. **Overlap-sensitive diagnosis.** We introduce overlap-restricted temporal variants and overlap-bucket analyses that reveal the temporal shift is *structured*: different train–test overlap strata favor different model families, explaining why aggregate rankings are unstable.
4. **Reusable benchmark resources.** We release split manifests with stable `example_id` keys and a standalone evaluator, enabling external models to be scored on the exact partitions used in this study.

2 Related Work

2.1 Cold-start evaluation in DTI

The evaluation of DTI models under cold-start conditions was formalized by Pahikkala et al. [14], who defined four settings—S1 (warm), S2 (drug-cold), S3 (target-cold), and S4 (double-cold)—and showed that performance drops sharply from S1 to S4. This taxonomy is now standard: most recent DTI methods report at least one cold-start variant [1, 9, 12, 23]. However, published evaluations typically optimize within a single benchmark design. Wang et al. [20] provided an early demonstration that the winner can change across settings—DeepDTI led in warm-start while Random Forest won in cold-drug—but this observation was incidental rather than the focus of a controlled benchmark study.

2.2 Split design and data leakage in molecular machine learning

The choice of train–test split has long been recognized as consequential in chemoinformatics. Sheridan [17] showed that time-split cross-validation produces error estimates much closer to true prospective performance than random splits. Yang et al. [22] evaluated 850+ experiments across 19 public and 16 proprietary datasets and found that scaffold splits better approximated temporal splits than random splits, while model rankings could shift between split types. More recently, Graber et al. [5] demonstrated that train–test data leakage in PDBbind had *severely* inflated deep-learning binding-affinity metrics; on their CleanSplit, apparent model superiority vanished. Schröder and Schwaller [15] proposed optimization-based splitting to minimize information leakage, and Landrein and Schrier [8] showed that scaffold splits still permit high molecular similarity between train and test.

These findings establish that split design matters, but they focus on molecular property prediction or binding affinity regression rather than on binary DTI classification under cold-start versus temporal shifts.

2.3 Temporal and prospective validation

Temporal evaluation—training on past data and testing on future data—is regarded as the gold standard for estimating prospective performance. Sheridan [17] introduced this idea for QSAR at Merck, and Janela and Bajorath [7] later proposed an algorithm to simulate time splits on public datasets based on Novartis lead-optimization projects. The Therapeutics Data Commons provides a DTI domain-generalization benchmark with patent-based temporal splits [6], where out-of-distribution performance degrades substantially. However, no prior study has directly compared the model rankings produced by synthetic cold-start splits with those from temporal/provenance-shifted evaluation on the same model panel and dataset family.

2.4 Benchmark sensitivity across molecular ML

Several large-scale studies have demonstrated benchmark sensitivity beyond DTI. Wu et al. [21] established MoleculeNet but its rankings are now considered unreliable due to dataset quality issues. Luo et al. [11] trained 62,820 models and found that "discrepancies in the actual data split across the literature can entail unfair performance comparison." Zheng et al. [24] showed substantial sensitivity to dataset quality, featurization, and splitting in small-molecule benchmarks. Ektefaie et al. [4] showed that in-distribution performance is a poor proxy for OOD performance across molecular tasks. Tossou et al. [19] found that SOTA models "do not work well consistently" on novel realistic endpoints and that classical descriptors sometimes outperform learned representations.

2.5 Similarity-aware and structure-enriched evaluation

Our overlap-bucket diagnosis is conceptually related to similarity-aware evaluation methods that enforce specific similarity distributions between train and test sets [8, 15]. The key difference is that we do not *design* the split to enforce a similarity

distribution; instead, we *diagnose* the naturally occurring similarity structure of temporal/provenance splits and show that it drives model-ranking changes. Recent structure-enriched DTI methods such as FusionDTI and SiamDTI incorporate 3D structural information or contrastive alignment that may behave differently in low-overlap regimes; evaluating such models on our benchmark suite is a natural extension.

2.6 Positioning of this work

The general principle that benchmark design affects model ranking is well established. Our contribution is to *instantiate and quantify* this principle for DTI prediction in a way that has not been done before: (a) we directly compare synthetic cold-start and temporal/provenance rankings on the same seven-model panel; (b) we separate temporal shift from provenance confounding using paired patent and non-patent benchmarks; and (c) we introduce overlap-sensitive diagnosis that reveals the temporal shift is structured rather than monolithic. The emphasis is on validation methodology rather than on proposing a new architecture.

3 Benchmark Setup

3.1 Model panel

We deliberately selected seven models spanning the major architectural families in DTI to ensure that benchmark sensitivity is not confined to one model type:

- `base` — descriptor-based MLP. Drug: RDKit Morgan fingerprint (2048 bits). Target: hashed protein k-mer vector (2048 dims). Pair feature: $[d, t, d \odot t, |d - t|]$, fed to a 2-layer MLP head. This is the anchor system for all delta comparisons; its full specification is in Appendix Table 7.
- `RF` — Random Forest on the *same* Morgan+k-mer descriptors as `base`, isolating the effect of the learning algorithm from the representation.

- `DTI-LM` — pooled ChemBERTa + ESM-2 features with a downstream MLP [1]. Represents the language-model pooling family.
- `HyperPCM` — task-conditioned variant where the target branch generates conditioning parameters (γ, β, weight, bias) for the drug representation. Represents the hypernetwork/conditioning family.
- `GraphDTA-style` — molecular GCN encoder + amino-acid CNN encoder [13]. Represents the graph-based family.
- `DrugBAN` — bilinear attention network that computes explicit drug-substructure $\times$ protein-subsequence interactions [2]. Represents the cross-attention interaction family.
- `ColdstartCPI-style` — per-atom GCN drug features and per-residue CNN protein features concatenated with learnable global tokens and processed by a Transformer self-attention layer [23]. Represents the Transformer-fusion family specifically designed for cold-start generalization.

This panel is not exhaustive—important architectures such as SP-DTI [9] are not included—but it covers seven distinct architectural families including a model specifically designed for cold-start scenarios, which is sufficient to demonstrate that benchmark sensitivity is a cross-family phenomenon.

Training fairness. To ensure that ranking differences reflect benchmark semantics rather than tuning asymmetry, we applied a uniform optimization protocol across all neural models: AdamW with learning rate 10^{-3}, weight decay 10^{-4}, BCE-with-logits loss with class-balanced positive weighting, and early stopping by validation AUPRC within a fixed epoch budget (6–8 epochs). Hyperparameters were not individually tuned per model–benchmark combination; each architecture uses a single configuration across all benchmarks (Appendix Tables 7 and 9). For `DTI-LM` and `HyperPCM`, the pretrained language-model encoders (ChemBERTa and ESM-2) are frozen; only the downstream MLP is trained. `RF` uses `scikit-learn` defaults with 400 estimators and

balanced class weighting. This design choice sacrifices per-model peak performance in favor of commensurable tuning effort, which is the appropriate trade-off for a study whose conclusion is about ranking stability rather than absolute performance. The reported rankings should therefore be interpreted as reflecting a fixed-tuning-budget regime; individual models might achieve higher absolute performance under dedicated tuning. In particular, language-model and cross-attention approaches may benefit from encoder fine-tuning or larger epoch budgets that our protocol does not provide. However, the ranking instability across benchmark axes is unlikely to vanish under richer tuning, because the reversals arise from benchmark-level distributional shift (e.g., drug-cold versus target-cold extrapolation demands) rather than from underfitting of a particular model family. We encourage future work to test this conjecture with per-model optimized configurations.

3.2 Benchmark taxonomy

We organize benchmarks into three families (Table 1):

Synthetic cold-start (`BindingDB_Kd`). Three split definitions—`unseen_drug` (S2 in the Pahikkala taxonomy), `unseen_target` (S3), and `blind_start` (S4, both drug and target unseen)—each removing different forms of entity overlap. These are the standard evaluation settings used in most recent DTI papers.

Temporal/provenance. Two year-based BindingDB benchmarks: a *patent* split (`patent_temporal`; train 2013–2017, validation 2018, test 2019–2021; 142K/41K/49K rows), and a *non-patent* split (`nonpatent_temporal`; train 1977–2018, validation 2019, test 2020–2023; 36K/3K/7K rows), constructed from the official BindingDB release after excluding patent-derived entries. The non-patent benchmark is critical because it controls for the patent-provenance confound: if rankings change only on the patent benchmark, the effect might be driven by patent curation rather than temporal shift

per se. We describe both as "temporal/provenance OOD" because even the non-patent benchmark mixes time shift with changes in assay composition and journal sourcing.

Supporting target-cold OOD. `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` provide shifted target-cold settings that test stability across assay and dataset boundaries.

Labeling and deduplication. All benchmarks use binary labels derived from a $\mathrm{p}K_d \geq 7.0$ threshold (equivalently, $K_d \leq 100\,\mathrm{nM}$). For standard `BindingDB_Kd` and `Ki` splits, raw affinities are converted via $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$; for patent-derived data, a patent-specific log-scale conversion is applied (Appendix Table 4). When multiple measurements exist for the same drug–target pair, the TDC preprocessing retains one representative value. Exact duplicates (same SMILES, same UniProt ID, same affinity) are removed before splitting; near-duplicates (salt/tautomer variants) are not canonicalized beyond RDKit's default SMILES canonicalization. Homologous-target filtering is not applied, so targets with high sequence similarity may appear in both train and test under entity-holdout splits. These choices are standard in the TDC framework but may affect absolute performance levels; we document them explicitly so that users can assess their impact.

Within the temporal family, we further construct two *overlap-restricted variants*: `patent_temporal_pair_novel` (test pairs whose exact drug–target combination is absent from training) and `patent_temporal_drug_novel` (test pairs whose drug scaffold is absent from training). These variants allow us to test whether conclusions hold when entity or pair recurrence is explicitly suppressed.

3.3 Metrics and uncertainty

We report AUPRC as the primary metric and AUROC as a secondary check. AUPRC is emphasized because the central question concerns imbalanced ranking quality. Because prevalence ranges from 0.06 (`DAVIS`) to 0.57 (patent), absolute AUPRC is compared only

Table 1 Benchmark taxonomy and seed-0 split statistics. Prevalence differs substantially across benchmarks, so AUPRC should be compared within benchmark, not across rows.

Benchmark	Shift family	Synth.?	Test rows	Pos. rate
`Kd / unseen_drug`	drug-cold	yes	12,744	0.1966
`Kd / blind_start`	drug+target-cold	yes	2,821	0.2031
`Kd / unseen_target`	target-cold	yes	12,052	0.2239
Patent temporal/prov.	temporal/prov.	no	49,028	0.5745
Nonpatent temporal/prov.	temporal/prov.	no	7,174	0.4260
`Ki / unseen_target`	assay+target-cold	mixed	70,380	0.4652
`DAVIS / unseen_target`	dataset+target-cold	mixed	5,168	0.0615

within a benchmark. All benchmarks use matched five-seed evaluation. Paired bootstrap confidence intervals (10,000 resamples) support the principal ranking comparisons; Appendix Table 12 collects these intervals.

3.4 Overlap diagnosis

For diagnostic analysis, each test example carries a saved `overlap_score` defined as $\min(s_d, s_t)$, where s_d is the cosine similarity between the test drug's 2048-bit Morgan fingerprint (radius 2) and its nearest training-set drug neighbor, and s_t is the cosine similarity between the test target's 2048-dimensional hashed k-mer vector and its nearest training-set target neighbor. These are the same descriptor-level representations used by `base` and `RF`; the overlap diagnostic therefore measures "support from the descriptor perspective," not from the LM or graph perspective. We use the minimum to ensure that a pair is labeled "high-overlap" only when *both* its drug and target are well represented in training; this is more conservative than using either side alone. Cosine similarity is used instead of Tanimoto so that drug-side and target-side scores share the same $[0, 1]$ scale within the joint diagnostic. The released overlap scores and per-sample predictions allow users to recompute diagnostics under alternative similarity definitions (e.g., LM-embedding cosine or structure-based similarity). We partition test examples into three equal-sized overlap buckets and report per-bucket performance. This diagnosis reveals *which overlap regime* drives the aggregate ranking, and whether different model families dominate in different strata.

3.5 Reusable resources

Each split is exported as a canonical bundle. The release includes: (i) split manifests with versioned metadata, split counts, and prevalence; (ii) partition tables (`train.csv`, `test.csv`) with stable `example_id` keys and exported overlap scores; (iii) entity tables with unique drug and target vocabularies; (iv) per-model prediction CSVs with per-sample probabilities; (v) model configuration files and training seeds; and (vi) a standalone evaluator (`scripts/evaluate_prediction_csv.py`) that scores external predictions against the exact test partitions used in this study. Appendix Table 14 lists the artifacts. A DOI-backed archive will be deposited at acceptance.

4 Results

4.1 Even synthetic cold-start rankings disagree

We begin with the `BindingDB_Kd` synthetic panel (Table 2). The key finding is that no single model dominates all three cold-start settings. On `unseen_drug`, `RF` achieves the highest mean AUPRC (0.7325), substantially ahead of all neural models. On `blind_start`, `RF` again leads (0.4335), closely followed by `base` (0.4296). However, on `unseen_target`, `base` is strongest (0.5739) and `HyperPCM` (0.5452) surpasses `RF` (0.5308). Notably, `ColdstartCPI-style`—a model specifically designed for cold-start generalization—ranks last or near-last on all three synthetic settings (0.4557, 0.3591, 0.3428), demonstrating that cold-start specialization does not guarantee strong cold-start performance across all split definitions.

This already challenges the common practice of reporting a single cold-start result: the winner on drug-cold is not the winner on target-cold, even within the same dataset. The effect spans all seven architectures and reflects the different demands that drug-side versus target-side extrapolation place on the representation and the learner.

A scaffold-restricted control sharpens this further. When entity holdout is replaced with a Murcko-scaffold split (`scaffold_drug`), the earlier `HyperPCM` edge over `base` on the drug side vanishes: the paired bootstrap CI for `HyperPCM` − `base` crosses zero (Appendix Table 11). The synthetic panel is useful, but its apparent best model depends on how chemically strict the split is.

4.2 Temporal/provenance benchmarks change the ranking

We next evaluate the same panel on the primary temporal/provenance benchmarks (Table 2, last two columns). On `patent_temporal`, `RF` achieves the highest AUPRC (0.8240), followed by `DTI-LM` (0.7850) and `base` (0.7770). `ColdstartCPI-style` (0.6285) ranks last, far below all other models. The key observation is *not* that `RF` wins—it is that the ranking differs from the synthetic panel (where `base` led on target-cold) and that the ordering among neural models also shifts (`DTI-LM` now outperforms `HyperPCM`, reversing the blind-start pattern).

The non-patent temporal benchmark strengthens this conclusion. On `nonpatent_temporal`, `RF` again leads (0.7201), with `DTI-LM` (0.6437) and `GraphDTA-style` (0.6289) above `base` (0.6348). Because this benchmark excludes patent-derived data, the ranking change cannot be attributed solely to patent-provenance curation. The temporal/provenance shift genuinely produces different model orderings from the synthetic cold-start panel.

4.3 Overlap policy within the temporal family matters

Even within the patent temporal family, the winner is sensitive to how much entity or pair recurrence is allowed (Appendix Table 11). In the `pair_novel` variant—where all test drug–target pairs are absent from training—`DTI-LM` becomes the AUPRC leader (0.7491), overtaking `RF` (0.7393), and the paired bootstrap CI for `DTI-LM` − `RF` is strictly positive. In the `drug_novel` variant, `DTI-LM` again leads (0.7402 vs. 0.7356 for `RF`).

Table 2 Main results: synthetic reference panel and primary temporal/provenance benchmarks. Each cell shows two lines: the *upper line* is AUPRC (mean ± std over five seeds) and the *lower line* is AUROC. Best mean AUPRC per column is bold. ColdstartCPI-style uses 4 seeds on blind-start and target-cold.

Model	BindingDB$_{Kd}$ drug-cold	BindingDB$_{Kd}$ blind-start	BindingDB$_{Kd}$ target-cold	Patent temporal/prov.	Nonpatent temporal/prov.
	0.6004 ± 0.0576	0.4296 ± 0.0770	**0.5739 ± 0.0778**	0.7770 ± 0.0072	0.6348 ± 0.0141
`base`	0.8209 ± 0.0184	0.7184 ± 0.0340	0.8076 ± 0.0416	0.7234 ± 0.0094	0.7337 ± 0.0065
	0.7325 ± 0.0216	**0.4335 ± 0.0528**	0.5308 ± 0.0607	**0.8240 ± 0.0012**	**0.7201 ± 0.0013**
`RF`	**0.8874 ± 0.0079**	**0.7375 ± 0.0274**	0.7949 ± 0.0280	**0.7778 ± 0.0016**	**0.7631 ± 0.0018**
	0.5268 ± 0.0513	0.3510 ± 0.0802	0.5300 ± 0.0818	0.7850 ± 0.0052	0.6437 ± 0.0372
`DTI-LM`	0.7670 ± 0.0262	0.6561 ± 0.0430	**0.8072 ± 0.0227**	0.7394 ± 0.0041	0.7539 ± 0.0172
	0.5689 ± 0.0562	0.3934 ± 0.1131	0.5452 ± 0.0853	0.7644 ± 0.0089	0.6189 ± 0.0314
`HyperPCM`	0.7963 ± 0.0330	0.7107 ± 0.0376	0.8005 ± 0.0312	0.7247 ± 0.0089	0.7366 ± 0.0192
	0.5813 ± 0.0312	0.4094 ± 0.0786	0.4853 ± 0.0946	0.7729 ± 0.0028	0.6289 ± 0.0243
`GraphDTA-style`	0.8175 ± 0.0190	0.7231 ± 0.0376	0.7680 ± 0.0422	0.7227 ± 0.0040	0.7195 ± 0.0102
	0.5886 ± 0.0209	0.4120 ± 0.0639	0.4521 ± 0.0745	0.7568 ± 0.0102	0.6156 ± 0.0214
`DrugBAN`	0.8269 ± 0.0196	0.7271 ± 0.0274	0.7497 ± 0.0274	0.7037 ± 0.0110	0.7053 ± 0.0117
	0.4557 ± 0.1174	0.3591 ± 0.0670	0.3428 ± 0.0959	0.6285 ± 0.0651	0.4678 ± 0.0578
`ColdstartCPI-style`	0.7249 ± 0.0843	0.6718 ± 0.0284	0.6210 ± 0.1314	0.5710 ± 0.0755	0.5576 ± 0.0748

This is an important finding: the identity of the best model changes not only across benchmark families but also across overlap policies *within* a single temporal benchmark. It implies that aggregate temporal rankings can be dominated by the easy, high-overlap portion of the test set, and that the model best suited for genuinely novel predictions may differ from the aggregate winner.

An earlier-cutoff probe (`patent_temporal_v2017`) preserves the `DTI-LM` > `base` ordering (Appendix Table 5), indicating that the effect is not an artifact of a particular year boundary.

4.4 The temporal shift is structured, not monolithic

To understand *why* temporal rankings differ from synthetic ones, we diagnose the patent benchmark by year band and overlap bucket (Table 3, Figure 2).

Year bands. `RF` leads in both the large 2019 slice and the pooled 2020–2021 slice. The temporal shift does not concentrate in one particular time period.

Overlap buckets. This is the most revealing analysis. In the *lowest-overlap* bucket (test examples least similar to training), `DTI-LM` is the strongest model (0.6353), ahead of `RF` (0.6005). But in the *middle-* and *high-overlap* buckets, `RF` dominates (0.7606 and 0.9426). Because the high-overlap stratum is large, `RF` wins the aggregate benchmark. However, a practitioner interested in truly novel predictions—the low-overlap regime—would select `DTI-LM` instead.

This structured pattern explains the instability: aggregate rankings are a mixture over heterogeneous difficulty strata, and different models specialize in different strata. The temporal shift is not simply "harder cold-start." It changes the composition of easy and hard examples, which changes which model's strengths matter most.

Table 3 Patent temporal diagnosis by year band and overlap bucket (AUPRC, mean ± seed std). Only the three models that define the aggregate and slice-level ranking changes are shown; the full panel is in Table 2. The 2021 slice (72 examples) is pooled into 2020–2021.

Slice type	Slice	Count	base	RF	DTI-LM
Year	2019	42,009	0.7626 ± 0.0098	**0.8062 ± 0.0013**	0.7734 ± 0.0071
Year	2020–2021	7,019	0.8716 ± 0.0121	**0.9098 ± 0.0005**	0.8533 ± 0.0100
Overlap	0 (0.00–0.53)	16,341	0.5551 ± 0.0190	0.6005 ± 0.0034	**0.6353 ± 0.0175**
Overlap	1 (0.53–0.71)	16,342	0.7218 ± 0.0167	**0.7606 ± 0.0012**	0.7489 ± 0.0094
Overlap	2 (0.71–0.99)	16,345	0.9237 ± 0.0030	**0.9426 ± 0.0005**	0.8866 ± 0.0097

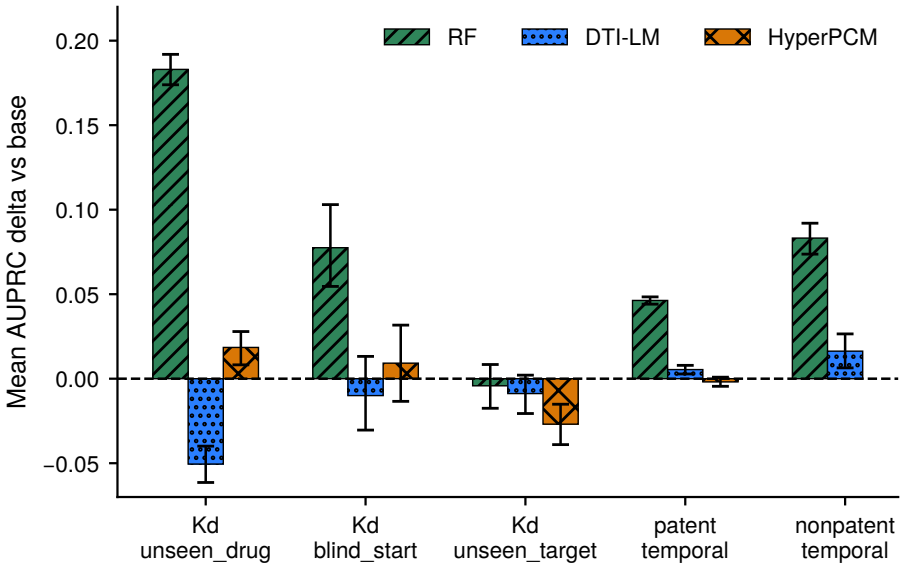

Fig. 1 AUPRC delta relative to `base` across all five primary benchmarks. `base` defines the zero line. Prevalence differs across benchmarks, so the vertical axis should be read as a within-benchmark contrast. Full absolute values are in Table 2.

4.5 Supporting benchmarks confirm heterogeneity

`BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` provide external checks on ranking stability (Appendix Table 10). On `Ki`, `base` leads (0.6934), ahead of all other models including `DTI-LM` (0.6648), `RF` (0.6328), and `DrugBAN` (0.6121). On `DAVIS`, `RF` regains the lead (0.4408), while `DrugBAN` (0.3164) falls to the bottom. Neither benchmark recovers a stable winner, reinforcing the main finding. These supporting benchmarks should not be over-interpreted as direct evidence for the temporal/provenance argument, since they differ from the main `BindingDB_Kd` setting in

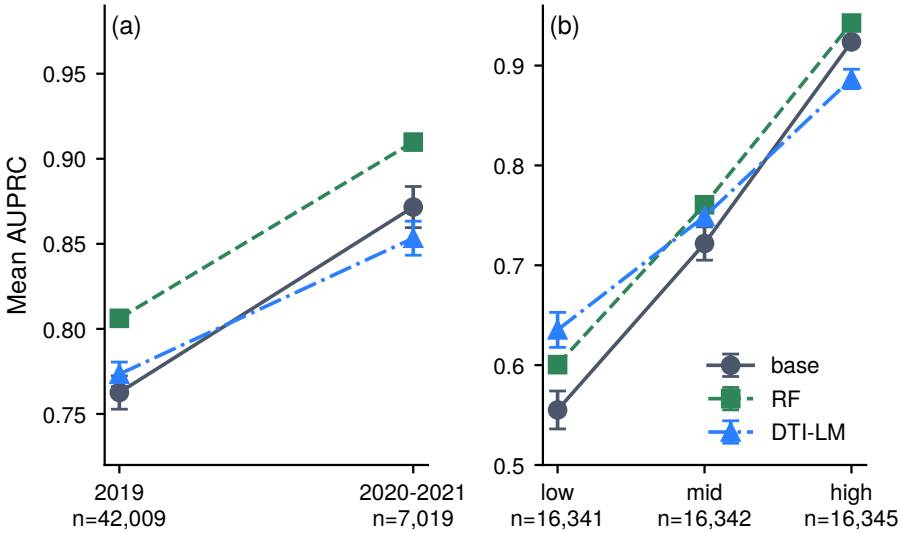

Fig. 2 Patent temporal diagnosis. (a) Year-band AUPRC for `base`, `RF`, and `DTI-LM`. (b) Overlap-bucket AUPRC. `RF` dominates aggregate and high-overlap strata; `DTI-LM` leads only in the lowest-overlap bucket. Error bars: seed std.

assay type and dataset source; their role is to demonstrate that ranking instability extends beyond the primary benchmark family.

Metric robustness. The ranking instability documented above is not an artifact of the AUPRC metric. Across the five primary benchmarks, the identity of the best model under AUROC agrees with AUPRC in four of five cases (Table 2). A Kendall's W rank concordance analysis confirms low cross-benchmark agreement under both threshold-free metrics: $W = 0.378$ (AUPRC) and $W = 0.374$ (AUROC) across all five benchmarks (Appendix Section I). The ranking non-transferability conclusion is therefore supported by two independent threshold-free metrics. Screening-oriented metrics such as enrichment factor or BEDROC would add useful context for low-prevalence benchmarks like `DAVIS`, but these require threshold or weighting choices that introduce additional benchmarking decisions; we leave them to future extensions.

5 Discussion

5.1 Implications for DTI model selection

The central message is straightforward: a single synthetic cold-start split is not sufficient evidence for DTI generalization. Our seven-model panel shows that the winner changes across (a) cold-start regimes, (b) synthetic versus temporal/provenance evaluation, and (c) overlap policies within a temporal benchmark. For CADD practitioners, the results translate into scenario-specific guidance:

- **Hit triage and lead expansion** (new compounds against known targets): the drug-cold regime is most relevant. Here `RF` and descriptor-based models dominate because held-out compounds remain structurally close to training scaffolds, so local interpolation suffices. A temporal/provenance benchmark should be checked in parallel, because patent-era compounds may introduce provenance-driven ranking shifts.
- **Target deconvolution and target-family extrapolation** (known compounds against new or distant targets): the target-cold regime applies, where `base` and language-model baselines outperform `RF`. In this regime, the overlap-bucket diagnosis shows that the low-overlap slice—where novel target predictions actually matter—favors `DTI-LM`, even though the aggregate benchmark is won by a different model. Practitioners should evaluate models on the low-overlap stratum rather than the aggregate.
- **Prospective virtual screening under temporal shift**: the patent temporal benchmark is the closest proxy. The aggregate winner (`RF`) benefits from high-overlap pairs that recur across years; when pair or drug novelty is enforced, `DTI-LM` overtakes `RF`. This means that for genuinely novel screening campaigns, the aggregate temporal leaderboard may overstate tree-ensemble performance.

Three reporting recommendations follow:

1. DTI studies should report at least one synthetic *and* one temporal/provenance-shifted benchmark when making generalization claims.
2. Overlap characteristics should be disclosed, because entity or pair recurrence can change the apparent ranking even within one benchmark family.
3. Claims about the superiority of any architecture family should be phrased in benchmark-qualified terms: "best under drug-cold holdout" is not the same as "best for novel-target screening."

5.2 Why Random Forest frequently wins

An unexpected empirical pattern deserves interpretation. `RF` leads on synthetic `unseen_drug`, `blind_start`, and both temporal/provenance panels, yet trails on `unseen_target` and `Ki`. Because `RF` and `base` use identical descriptors, the pattern is not about representation. We interpret it as *descriptor-space interpolation under drug-side support*: when held-out drugs remain similar to training scaffolds (median nearest-train Tanimoto 0.77 on `unseen_drug`), tree ensembles exploit local descriptor neighborhoods effectively. When the dominant challenge shifts to protein-side extrapolation, the hashed k-mer representation becomes the bottleneck and the `RF` advantage vanishes.

This interpretation is consistent with the overlap-bucket diagnosis: `RF` dominates the high-overlap strata where interpolation is easiest, while `DTI-LM`'s learned representations become advantageous in the low-overlap regime. The recurrent `RF` gains should therefore not be read as "classical beats deep learning" but as evidence that descriptor-level interpolation suffices when drug-side support is high—a regime-specific finding rather than a universal one.

5.3 Limitations

ColdstartCPI-style fidelity. Our `ColdstartCPI-style` model is a simplified reimplementation that preserves the core architectural contribution of Zhao et al.

[23]—Transformer self-attention over concatenated per-atom drug and per-residue protein tokens—but differs from the original in two ways: it uses GCN-derived atom features instead of Mol2Vec embeddings, and CNN-derived residue features instead of ProtTrans-BERT-BFD. It also uses the same fixed-tuning-budget protocol as all other models (6–8 epochs, AdamW with lr 10^{-3}). The original ColdstartCPI reports substantially higher numbers on its own BindingDB splits; therefore, the low absolute performance of `ColdstartCPI-style` in our study likely reflects encoder and tuning differences rather than a fundamental limitation of the Transformer-fusion architecture. For the ranking-stability argument, this is acceptable: the model still represents a genuinely different fusion strategy, and its ranking *relative to other models* varies across benchmarks (e.g., it is closer to the pack on patent temporal than on target-cold synthetic). A fairer head-to-head evaluation of the full ColdstartCPI pipeline with its native encoders is left to future work.

Panel scope. Seven models spanning seven architectural families provide sufficient evidence for ranking instability, but the panel is not exhaustive. Recent token-level fusion methods (e.g., FusionDTI), contrastive approaches (e.g., SiamDTI), and 3D-structure-aware models (e.g., SP-DTI [9]) are not included. Structure-enriched models may behave differently in the low-overlap regime where LM-based approaches currently lead; testing this is a natural extension. A larger community benchmark would be needed to quantify how far the ranking-instability effect extends across the full modern DTI literature.

Confounded temporal shift. The patent benchmark mixes temporal, provenance, assay, and possibly label-noise shifts. We do not claim to isolate which factor drives the ranking change; our claim is that the *aggregate* temporal/provenance evaluation produces different rankings from synthetic cold-start, which is the practically relevant comparison for model selection. The non-patent benchmark partially controls for provenance and preserves the ranking change, but it is smaller (7,174 vs. 49,028 rows).

A cleaner causal attribution would require source-matched temporal cohorts with harmonized assay composition.

Seed count and metrics. Five seeds provide reasonable coverage but high-variance settings (e.g., synthetic `unseen_drug`) could benefit from even more seeds. We report only threshold-free metrics (AUPRC and AUROC). For low-prevalence settings such as `DAVIS` ($\sim$6% positive), screening-oriented metrics including enrichment factor at fixed false-positive rates, BEDROC, or top-k hit rate would be directly relevant to virtual screening campaigns. These were omitted to keep all panels on one metric pair, but the released per-sample predictions enable users to compute any desired metric post hoc.

Binarization threshold. All benchmarks use $\mathrm{p}K_d \geq 7.0$ as the activity threshold. Different thresholds would change class prevalence and could shift rankings; we did not sweep this parameter, and sensitivity analysis across thresholds is left to future work.

No prospective validation. We do not evaluate biological decision quality. The results demonstrate ranking instability under retrospective evaluation designs, not prospective predictive value.

6 Conclusion

We presented a controlled benchmark study showing that model rankings in DTI prediction are not preserved across evaluation axes. Across a seven-model panel—including `ColdstartCPI-style`, a model specifically designed for cold-start generalization—synthetic cold-start splits already produced different winners depending on which entity is held out. Temporal/provenance benchmarks changed the ranking further, and this effect survived on non-patent data, ruling out a patent-provenance artifact. Within the temporal family, overlap-restricted variants and overlap-bucket diagnosis revealed that the shift is structured: different overlap strata favor different model families. These findings establish that synthetic cold-start and temporal/provenance benchmarks are non-interchangeable validation settings for DTI. Future studies should

report both axes when making generalization claims, and benchmark-qualified language should replace architecture-level universal claims.

Statements and Declarations

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Competing interests The authors declare that they have no known financial or non-financial competing interests relevant to this study.

Author contributions Author names and contribution details will be finalized at submission.

Data availability A fixed review snapshot corresponding to git commit `a5d6b8b` (`a5d6b8bd96c98fed73c5a9771c2a1079c9f12255`) is archived with the submission and mirrored in the project repository at https://github.com/cicisoso/Synthetic-Cold-Splits-Are-Not-Enough. The snapshot includes the exported benchmark resources under `benchmark_resources/`, the benchmark tables under `reports/`, and the release manifest at `benchmark_resources/RELEASE_MANIFEST.md`. A DOI-backed public archive will be deposited at acceptance.

Code availability The exact code used to construct the benchmark splits, train the comparison models, and reproduce the reported analyses corresponds to commit `a5d6b8b`. The evaluator entry points are `scripts/evaluate_prediction_csv.py` and `scripts/evaluate_decision_csv.py`; the full release contents are enumerated in `benchmark_resources/RELEASE_MANIFEST.md`.

Table 4 Specification of the primary patent temporal/provenance benchmarks. The year partition is fixed; the random seed changes training stochasticity only.

Aspect	`patent_temporal`	`patent_temporal_v2017`
Source files	`train_val.csv` + `test.csv`	`train_val.csv` + `test.csv`
Train years	2013–2017	2013–2016
Validation year	2018	2017
Test years	2019–2021	2018–2021
Rows (train / valid / test)	142,275 / 41,155 / 49,028	90,771 / 51,504 / 90,183
Pos. rate (train / valid / test)	0.566 / 0.534 / 0.575	0.568 / 0.562 / 0.556
Label transform	patent log-scale transform $Y \mapsto \mathrm{p}K_d = 9 - Y/\ln(10)$; positive if $\mathrm{p}K_d \geq 7.0$	
Overlap policy	entity overlap allowed; similarity exported through `overlap_score`	
Duplicate policy	pair rows retained after year split; entity tables deduplicated by canonical ID	
Seed semantics	year partition fixed; seed changes training stochasticity only	
OOD interpretation	temporal/provenance OOD, not pure time shift	

Table 5 Earlier-cutoff patent robustness probe (`patent_temporal_v2017`).

Benchmark	`base` AUPRC AUROC	`DTI-LM` AUPRC AUROC	Reading
`BindingDB_patent`	0.7044 ± 0.0058	0.7134 ± 0.0027	
`patent_temporal_v2017`	0.6624 ± 0.0079	0.6754 ± 0.0039	earlier cutoff preserves `DTI-LM > base`

Appendix A Patent benchmark specification

Table 4 enumerates the year boundaries, split counts, label construction, overlap policy, and seed semantics for both the main patent split and the earlier-cutoff robustness variant.

Appendix B Non-patent temporal probe specification

Table 6 lists the source, year cutoffs, label transform, prevalence, and seed semantics for the non-patent temporal split.

Table 6 Specification of the non-patent temporal probe.

Aspect	`nonpatent_temporal`
Source file	official BindingDB TSV, filtered to rows without a patent number
Date field	publication year from `Date of publication`
Train years	1977–2018
Validation year	2019
Test years	2020–2023
Rows (train / valid / test)	35,755 / 3,006 / 7,174
Pos. rate (train / valid / test)	0.341 / 0.445 / 0.426
Label transform	raw K_d in nM with $\mathrm{p}K_d = -\log_{10}(Y/10^9)$; positive if $\mathrm{p}K_d \geq 7.0$
Overlap policy	entity overlap allowed; `overlap_score` exported
Duplicate policy	exact duplicate rows removed before partitioning
Seed semantics	year partition fixed; seed changes training stochasticity only
OOD interpretation	non-patent temporal shift with less provenance mixing

Table 7 Specification of the `base` reference model used throughout the benchmark.

Component	Specification
Drug input	RDKit Morgan fingerprint, radius 2, 2048 bits.
Target input	L2-normalized hashed amino-acid 1/2/3-mer count vector, 2048 dimensions.
Drug encoder	Two-layer MLP: Linear$(2048, 512)$, GELU, dropout 0.1, Linear$(512, 256)$, L2 norm.
Target encoder	Same architecture as drug encoder, independent parameters.
Pair head	$[d,\ t,\ d \odot t,\ \lvert d - t \rvert] \in \mathbb{R}^{1024}$, then Linear$(1024, 512)$, GELU, dropout 0.1, Linear$(512, 1)$.
Parameters	2,886,145 total.
Optimization	BCE-with-logits, Adam (lr 10^{-3}, wd 10^{-4}), 8 epochs, batch 256.
Labeling	Binary from $\mathrm{p}K_d \geq 7.0$ (patent: log-nM conversion per Table 4).
Code path	`src/raicd/model.py`; training: `scripts/train_raicd.py`.

Table 8 Specification of `DTI-LM` and `HyperPCM` baselines.

Component	Specification
DTI-LM drug input	Mean-pooled ChemBERTa (DeepChem `seyonec/ChemBERTa-zinc-base-v1`) last hidden state, 768-d. Encoder frozen.
DTI-LM target input	Mean-pooled ESM-2 (`esm2_t12_35M_UR50D`) last hidden state, 480-d. Encoder frozen.
DTI-LM pair head	Concatenate $[d, t] \in \mathbb{R}^{1248}$, then Linear$(1248, 512)$, ReLU, dropout 0.1, Linear$(512, 256)$, ReLU, dropout 0.1, Linear$(256, 1)$.
DTI-LM optimization	BCE-with-logits, AdamW (lr 10^{-3}, wd 10^{-4}), 6 epochs, batch 128.
DTI-LM code path	`src/raicd/lm_baselines.py`; training: `scripts/train_recent_baseline.py --model dtilm`.
HyperPCM architecture	Same LM encoders as DTI-LM (frozen). Target embedding generates conditioning parameters (γ, β, W, bias) via hypernetwork MLP; drug embedding is then FiLM-conditioned before prediction head.
HyperPCM optimization	Same as DTI-LM: BCE-with-logits, AdamW (lr 10^{-3}, wd 10^{-4}), 6 epochs, batch 128.
HyperPCM code path	`src/raicd/lm_baselines.py`; training: `scripts/train_recent_baseline.py --model hyperpcm`.

Appendix C Reference-model specification

Appendix D Language-model baseline specifications

Appendix E Random Forest, graph-based, Transformer-fusion, and cross-attention baseline specifications

Appendix F Additional real-OOD support benchmarks

Appendix G Additional robustness analyses

Appendix H Paired uncertainty support

Appendix I Rank concordance analysis

To quantify ranking stability globally rather than through pairwise comparisons alone, we compute Kendall's coefficient of concordance (W) across the five primary benchmarks

Table 9 Specification of the `RF`, `GraphDTA-style`, and `DrugBAN` baselines.

Component	Specification
GraphDTA drug input	RDKit molecular graph. Node features: atom symbol, degree, H count, valence, charge, aromaticity, hybridization.
GraphDTA drug encoder	Three-layer GCN ($36 \to 128 \to 128 \to 128$), ReLU, dropout 0.1. Global mean+max pool $\to$ 256-d.
GraphDTA target input	Amino-acid tokens (vocab 21), truncated/padded to 1000.
GraphDTA target encoder	Embedding 64-d, three Conv1d ($64 \to 128$, $k = 8$), ReLU, dropout 0.1, global max pool $\to$ 128-d.
GraphDTA pair head	Linear$(384, 256)$, ReLU, dropout 0.1, Linear$(256, 128)$, ReLU, dropout 0.1, Linear$(128, 1)$.
GraphDTA params	499,009 total. Optimization: BCE-with-logits, AdamW (lr 10^{-3}, wd 10^{-4}), 8 epochs, batch 128.
DrugBAN architecture	Bilinear attention network: GNN-encoded molecular substructure features interact with CNN-encoded protein subsequence features via a bilinear attention mechanism, producing explicit drug–protein interaction maps [2].
DrugBAN code path	Adapted from https://github.com/peizhenbai/DrugBAN.
ColdstartCPI-style architecture	Transformer self-attention over concatenated drug atom and protein residue tokens [23]. Drug encoder: same 3-layer GCN as GraphDTA, returning per-atom features (max 100 atoms). Protein encoder: same 3-layer CNN, returning per-residue features (max 200 residues). Two learnable global tokens (drug CLS, protein CLS) are prepended. Single `TransformerEncoderLayer` with $d_{\text{model}} = 128$, $n_{\text{heads}} = 2$, $d_{\text{ff}} = 512$, dropout 0.1. Pair head: $[d_{\text{CLS}}, p_{\text{CLS}}] \to$ 3-layer MLP with PReLU $\to$ 1.
ColdstartCPI-style note	Simplified reimplementation; uses GCN/CNN features instead of Mol2Vec/ProtTrans as in the original. See Limitations for fidelity discussion.
ColdstartCPI-style code path	`src/raicd/graph_baselines.py`; training: `scripts/train_graph_baseline.py --model coldstartcpi`.
RF hyperparams	`n_estimators`= 400, `max_depth`=None, `min_samples_leaf`= 1, `max_features`=`sqrt`, `class_weight`=`balanced_subsample`.
RF code path	`src/raicd/classical_baselines.py`; training: `scripts/train_classical_baseline.py`.

for both AUPRC and AUROC. We also report pairwise Spearman rank correlations between benchmark columns.

Ranks are computed from the mean metric value across five seeds for each model–benchmark pair; ties are resolved by midrank. Kendall's W is computed over the seven models ranked across either all five or a subset of benchmarks. The low W values (≈ 0.37–0.38) confirm that model rankings are not well preserved across benchmarks under either metric. Drug-cold and target-cold rankings are negatively correlated ($\rho = -0.26$ to -0.43), indicating that these two synthetic settings reward opposite model strengths. Drug-cold and patent temporal rankings are essentially independent ($\rho \approx 0$), while target-cold and patent temporal are positively correlated ($\rho = +0.77$

Table 10 Supporting real-OOD benchmarks (matched 5-seed where available, 3-seed for legacy models). Best mean AUPRC per row is bold.

Benchmark	base	RF	DTI-LM	HyperPCM	GraphDTA-style	DrugBAN
`Ki` / `unseen_target`	**0.6934 ± 0.0257** / 0.7192 ± 0.0068	0.6328 ± 0.0465 / 0.6581 ± 0.0319	0.6648 ± 0.0177 / 0.6936 ± 0.0044	0.6515 ± 0.0301 / 0.6757 ± 0.0211	0.6280 ± 0.0237 / 0.6550 ± 0.0193	0.6121 ± 0.0337 / 0.6306 ± 0.0294
`DAVIS` / `unseen_target`	0.4097 ± 0.0275 / 0.8613 ± 0.0153	**0.4408 ± 0.0325** / 0.8806 ± 0.0128	0.4090 ± 0.0570 / 0.8718 ± 0.0152	0.3840 ± 0.0269 / 0.8610 ± 0.0245	0.3601 ± 0.0502 / 0.8359 ± 0.0292	0.3164 ± 0.0524 / 0.7818 ± 0.0437

Table 11 Robustness analyses: provenance removal, scaffold holdout, and overlap-restricted patent variants. Values are mean ± std over three seeds.

Benchmark	Systems	Mean AUPRC / AUROC	Paired AUPRC CI	Read
`nonpatent_temporal`	`base` vs `DTI-LM`	`base`: 0.6369 ± 0.0173 / 0.7333 ± 0.0081; `DTI-LM`: **0.6531 ± 0.0066 / 0.7644 ± 0.0041**	+0.0163 [+0.0064, +0.0265]	ranking change survives without patent provenance
`scaffold_drug`	`base` vs `HyperPCM`	`base`: **0.6069 ± 0.0161 / 0.8351 ± 0.0037**; `HyperPCM`: 0.5983 ± 0.0368 / 0.8155 ± 0.0204	−0.0086 [−0.0203, +0.0025]	scaffold holdout removes `HyperPCM` edge
`patent_pair_novel`	`base`, `RF`, `DTI-LM`	`base`: 0.7378 ± 0.0107; `RF`: 0.7393 ± 0.0015; `DTI-LM`: **0.7491 ± 0.0076**	`DTI-LM - RF`: +0.0098 [+0.0061, +0.0134]	pair-novel flips leader from `RF` to `DTI-LM`
`patent_drug_novel`	`base`, `RF`, `DTI-LM`	`base`: 0.7255 ± 0.0117; `RF`: 0.7356 ± 0.0016; `DTI-LM`: **0.7402 ± 0.0077**	`DTI-LM - RF`: +0.0047 [+0.0009, +0.0083]	drug-novel still favors `DTI-LM`

Table 12 Paired bootstrap support for principal ranking comparisons (based on the seed-0 prediction set). Each row shows `A − B`; a positive delta means `A` outperforms `B`. Note: deltas here are sample-level paired bootstrap on a single seed; they may differ slightly from the 5-seed mean deltas in Table 2 because the table reports cross-seed averages.

Comparison	Mean AUPRC delta	95% CI
`unseen_drug`: `RF - base`	+0.1830	[+0.1739, +0.1919]
`unseen_drug`: `RF - HyperPCM`	+0.1645	[+0.1558, +0.1733]
`blind_start`: `RF - base`	+0.0775	[+0.0546, +0.1030]
`scaffold_drug`: `HyperPCM - base`	−0.0086	[−0.0203, +0.0025]
`unseen_target`: `RF - base`	−0.0042	[−0.0175, +0.0084]
`patent`: `RF - base`	+0.0463	[+0.0441, +0.0484]
`patent`: `RF - DTI-LM`	+0.0409	[+0.0384, +0.0433]
`patent_pair_novel`: `RF - base`	+0.0015	[−0.0022, +0.0049]
`patent_pair_novel`: `DTI-LM - RF`	+0.0098	[+0.0061, +0.0134]
`patent_drug_novel`: `RF - base`	+0.0101	[+0.0065, +0.0137]
`patent_drug_novel`: `DTI-LM - RF`	+0.0047	[+0.0009, +0.0083]
`nonpatent_temporal`: `DTI-LM - base`	+0.0163	[+0.0064, +0.0265]
`nonpatent_temporal`: `RF - base`	+0.0832	[+0.0737, +0.0920]
`nonpatent_temporal`: `RF - DTI-LM`	+0.0670	[+0.0564, +0.0764]

Table 13 Rank concordance analysis. Kendall's W ranges from 0 (no agreement) to 1 (perfect agreement). Spearman ρ is shown for selected benchmark pairs. All analyses use the seven-model panel.

Statistic	AUPRC	AUROC
Kendall's W (all 5 benchmarks)	0.378 ($p = 0.09$)	0.374 ($p = 0.10$)
Kendall's W (synthetic only, 3 benchmarks)	0.416	0.365
Kendall's W (temporal only, 2 benchmarks)	0.771	0.914
Spearman ρ: drug-cold vs. target-cold	−0.26	−0.43
Spearman ρ: drug-cold vs. patent temporal	−0.09	−0.03
Spearman ρ: target-cold vs. patent temporal	+0.83	+0.77

to +0.83), suggesting that temporal shift shares some structure with protein-side extrapolation. The within-temporal concordance is much higher ($W = 0.77$–0.91), indicating that the two temporal benchmarks agree with each other more than either agrees with drug-cold evaluation.

Table 14 Reusable benchmark artifacts released with each split.

Artifact	Example files	Purpose
Split manifest	`manifest.json`	versioned metadata, split counts, prevalence, seed, canonical paths
Partition tables	`train.csv`, `test.csv`	labeled partitions with `example_id` and overlap scores
Pair tables	`train_pairs.csv`	lightweight pair-only views for external pipelines
Entity tables	`drugs.csv` / `targets.csv`	unique drug and target vocabularies
Full joined table	`all.csv`	complete resource snapshot for auditing
Standalone evaluator	`evaluate_prediction_csv.py`	scores prediction CSV; returns global and overlap-bucket metrics

Appendix J Reusable benchmark resources

Appendix K Versioned release snapshot

The manuscript is tied to git commit `a5d6b8b`. The archived snapshot includes exported split bundles under `benchmark_resources/`, evaluators in `scripts/`, and summaries under `reports/`. The manifest `benchmark_resources/RELEASE_MANIFEST.md` records the full commit hash and artifact paths.

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